Extra Credit 2

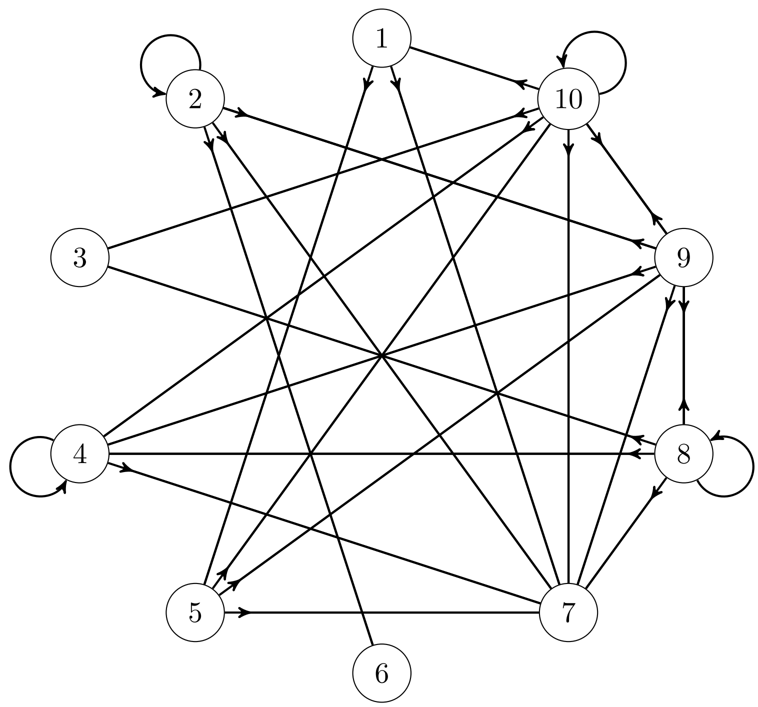
Due Date: September 6, 2019 by 5:00pm

MAT 022A Linear Algebra
Instructor: Mikhail Gaerlan

Instructions: This assignment is worth 15 extra points added to Midterm 2.

A small company runs an internal network of 10 websites, which has no links to the outside world. The links within the internal network are shown in the graph below. For example, website 1 has links to websites 5 and 7.

Use the PageRank algorithm below with the help of the Power method to rank the websites. Find the matrix A, the vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{10}$, and then rank the websites from most to least important.



PageRank algorithm:

1. Find a matrix A as follows:
   (a) For each website i, find n_i where n_i is the number of outgoing links from website i to a different website. Do not include links from a website to itself.
   (b) Start with a matrix $A = (a_{ij})$ where
   $$a_{ij} = \begin{cases} \dfrac{1}{n_i} & , \text{ if } i \neq j \text{ and website } i \text{ has a link to website } j \\ 0 & , \text{ otherwise} \end{cases}$$
   (c) If a row contains all zeros, replace all elements in that row with $\dfrac{1}{n}$ where n is the number of websites.

2. Find the eigenvector $A^T\boldsymbol{x} = \boldsymbol{x}$. Use the Power method to approximate $\boldsymbol{x}$ with $\boldsymbol{x}_0 = \mathbf{1}$ where $\mathbf{1}$ is the vector with all ones and $k = 10$.

   Power method:

   (a) For $i = 1, \ldots, k$

      i. Find $\boldsymbol{y}_i = A^T\boldsymbol{x}_{i-1}$ and set λ_i to be the largest value in $\boldsymbol{y}_i$.

      ii. Set $\boldsymbol{x}_i = \boldsymbol{y}_i/\lambda_i$.

   (b) $\boldsymbol{x}_k$ will the the approximation to the eigenvector $\boldsymbol{x}$.

3. Rank the websites by the values in the eigenvector. Each element of the eigenvector corresponds to a website, e.g. the fourth element of the eigenvector corresponds to website 4. The higher the value in the eigenvector, the more important the website is.

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} \\ \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} & \frac{1}{10} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 \\ 0 & \frac{1}{5} & 0 & \frac{1}{5} & 0 & 0 & \frac{1}{5} & \frac{1}{5} & 0 & \frac{1}{5} \\ \frac{1}{5} & 0 & \frac{1}{5} & \frac{1}{5} & 0 & 0 & \frac{1}{5} & 0 & \frac{1}{5} & 0 \end{bmatrix}$$

Ranking:
1. Website 7
2. Website 9
3. Website 4
4. Website 10
5. Website 3
6. Website 5
7. Website 2, Website 8
9. Website 6
10. Website 1

$$X = \left[\vec{x}_1 \ \vec{x}_2 \ \vec{x}_3 \ \vec{x}_4 \ \vec{x}_5 \ \vec{x}_6 \ \vec{x}_7 \ \vec{x}_8 \ \vec{x}_9 \ \vec{x}_{10}\right]$$

$$= \begin{bmatrix} 0.160428 & 0.231975 & 0.212520 & 0.215778 & 0.215190 & 0.215783 & 0.215295 & 0.215563 & 0.215438 & 0.215495 \\ 0.160428 & 0.275862 & 0.224882 & 0.244018 & 0.236159 & 0.240002 & 0.238032 & 0.238989 & 0.236539 & 0.238749 \\ 0.240642 & 0.278997 & 0.277276 & 0.272998 & 0.275946 & 0.274970 & 0.275190 & 0.275127 & 0.275159 & 0.275142 \\ 0.304813 & 0.385580 & 0.360283 & 0.363791 & 0.362466 & 0.363937 & 0.362994 & 0.363448 & 0.363242 & 0.363337 \\ 0.256684 & 0.263323 & 0.250785 & 0.261374 & 0.257088 & 0.258900 & 0.257929 & 0.258415 & 0.258192 & 0.258294 \\ 0.203209 & 0.231975 & 0.228218 & 0.229519 & 0.230647 & 0.229952 & 0.230087 & 0.230085 & 0.230086 & 0.230083 \\ 1.000000 & 1.000000 & 1.000000 & 1.000000 & 1.000000 & 1.000000 & 1.000000 & 1.000000 & 1.000000 & 1.000000 \\ 0.160428 & 0.275862 & 0.224882 & 0.244018 & 0.236159 & 0.240002 & 0.238032 & 0.238989 & 0.238539 & 0.238749 \\ 0.454545 & 0.442006 & 0.446036 & 0.434372 & 0.443724 & 0.439797 & 0.441198 & 0.440600 & 0.440890 & 0.440751 \\ 0.267380 & 0.376176 & 0.307300 & 0.329099 & 0.322929 & 0.325913 & 0.324180 & 0.325045 & 0.324640 & 0.324830 \end{bmatrix} \begin{array}{c} 10 \\ 7 \\ 5 \\ 3 \\ 6 \\ 9 \\ 1 \\ 7 \\ 2 \\ 4 \end{array}$$